

Shafiq-us Saleheen

Computer Science PhD Student | Purdue University

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Portfolio: <https://saleheenshafiq9.github.io/>

RESEARCH INTERESTS

Autonomous Systems Security, Software Vulnerability Discovery, and Program Analysis
(exploring fuzzing, binary-patch analysis and reverse engineering techniques)

EDUCATIONAL BACKGROUND

Purdue University - West Lafayette, Indiana, USA

PhD in Computer Science

Aug 2025

- PurSec Lab (Advised by Dr. Antonio Bianchi)

Institute of Information Technology, University of Dhaka, Bangladesh

B.Sc. in Software Engineering

Jan 2019 - Dec 2023

- CGPA: 3.86/4.00

IELTS

- Overall: 8
 - Listening: 8.5, Reading: 8.5, Speaking: 7, Writing: 7.5
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Jan 2025

EXPERIENCE

Graduate Research Assistant

Purdue University, West Lafayette, IN, USA

Jan 2026 - Present

- Currently working as an RA with Dr. Antonio Bianchi & Dr. Dongyan Xu
 - Working on vulnerability discovery frameworks for unmanned autonomous vehicles (UAVs)
 - Investigating on different forms of binary and patch analysis techniques
 - Developing time-sensitive architectures for Drone software

Graduate Teaching Assistant

Purdue University, West Lafayette, IN, USA

Aug 2025 - Dec 2025

- Worked as a GTA for CS251: *Data Structures & Algorithms* -
 - Leading a PSO (Practice Study Observation) session
 - Holding Office Hours
 - Involved in grading, proctoring, question-setting and proof-reading.

Software Engineer

Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

Feb 2024 - Jul 2025

- Samsung's SmartThings Web project -
 - Contributed as full-stack developer.
- Research & Innovation -
 - Submitted DOI, built PoCs for generative AI, UWB/BLE & AR/VR related solutions.
 - Training on system design fundamentals, clean code & refactoring, generative AI.
 - **Technical Skill:** NEXT JS, Micronaut (Kotlin), DynamoDB, Android SDK, Java, FastApi, Docker, MongoDB, MySQL, Ubuntu Linux, Git

Researcher Secure & Resilient Cyber-Physical Systems (SECPS) Lab Arizona State University, Tempe, AZ, USA	Dec 2024 - Jul 2025
Currently working under the supervision of Dr. Ozgur Ozmen	
Researcher HIVE (Human Signal in Intelligence, Verification & AccEssibility) Lab Pennsylvania State University, University Park, PA, USA	Feb 2025 - Aug 2025
Worked under the supervision of Dr. Tanusree Sharma	
Research Intern Secure Platforms Lab (SPL) William & Mary, Williamsburg, VA, USA	Jun 2023 - Dec 2023
Worked under the supervision of Dr. Adwait Nadkarni	
Software Engineering Intern Streams Tech Ltd.	Dec 2022 - Jun 2023
Technical Skill: C#, Python, Docker, PostgreSQL, QGIS, PostGIS	

RESEARCH PROJECTS

Automated Backporting and Fuzzing on Drone Softwares Purdue University, West Lafayette, IN, USA	Jan 2026 - Present
- Exploring automated backporting techniques to analyze binaries - Adding analysis-aware fuzzing infrastructure	
Attack on Drone Recovery Systems Arizona State University, Tempe, AZ, USA	Dec 2024 - Present
- Collaborated with Dr. Ozgur Ozmen from Arizona State University - Collaborated with Dr. Qingzhao Zhang from University of Arizona	
D2M: User-Controlled Data Minimization for AI Document Processing Pennsylvania State University, University Park, PA, USA	Feb 2025 - Aug 2025
Collaborated with Dr. Tanusree Sharma from Pennsylvania State University	
IoTWhiz: An IoT Android App Characterization Tool University of Dhaka, Dhaka, Bangladesh (In collab. with William & Mary, Williamsburg, VA, United States)	Jun 2023 - Dec 2023
- Developed a tool to characterize IoT/non-IoT android applications - Automated report generation, comparing IoT and non-IoT apps across various metrics.	
PharmaSync: Optimizing Security and Transparency in Pharmaceutical Supply Chains using Decentralized Hybrid Blockchain International Blockchain Olympiad 2023, Amsterdam, Netherlands	Jul 2023 - Present
- Developed to address critical issues in pharmaceutical supply chains. - Integrated zero-knowledge proofs to enhance security while maintaining transparency. - Implemented hybrid blockchain architecture to balance transparency and privacy. Paper Link: PDF	

ACHIEVEMENTS

Travel Grant, GREPSEC VII Workshop, 35th USENIX Security Symposium, Baltimore, MD, USA

- Awarded student travel grant to attend early-career researchers' workshop.

Award of Distinction, 6th International Blockchain Olympiad 2023, Amsterdam, Netherlands

- Won the award for a supply chain solution named - "PharmaSync: Optimizing Security and Transparency in Pharmaceutical Supply Chains using Decentralized Hybrid Blockchain"
- Certificate Link: [PDF](#)

Professor Jamilur Reza Choudhury Championship Award, 4th National Blockchain Olympiad 2023, Dhaka, Bangladesh

- News Link: [‘Team Pharmars’ -the Best team at the 4th Blockchain Olympiad](#)

Finalist, Learnathon 2022, Dhaka, Bangladesh

- Won the award for a twitter clone website using Angular & .NET framework

Finalist, Hult Prize 2023-24 on Campus Program of ULAB, Dhaka, Bangladesh

- Won the award for an idea called "Turning Invasive Suckerfish into Sustainable Fertilizer".

Champion, RUET CSE Fest 2k22, Rajshahi, Bangladesh

- Won the award for an app for chemistry learning called "Chemouflage" using Unity game engine.

SKILLS

- Programming Languages: C, C++, C#, Python, Java, Javascript, Kotlin, R, PHP, Dart, Solidity
- Front-end Technologies: React JS, Next JS, Angular, Tailwind CSS, Bootstrap, HTML, Flutter
- Back-end Technologies: FastApi, Django REST, Node JS, Express JS, Micronaut
- Database Technologies: MySQL, PostgreSQL, SQLite, DynamoDB, MongoDB
- Others: Ghidra, IDA-Pro, Docker, Git, Postman, Design Patterns, Microservice basics, Tableau

REFERENCES

Antonio Bianchi
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Purdue University, West Lafayette, IN, United States
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Kazi Muheymin-Us-Sakib
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